

Firmware 7.15 blocks 8–10: final integration audit packet

Block09 is complete by owner direction and its exact published head is integrated. This packet supersedes the provisional combined report. Block10 is ready for the final software audit after the local preflight recorded in the live receipt. No block10 hosted CI or external review result is claimed. Physical-device and release acceptance remain separate.

Identity and scope

Item	Immutable identity
Accepted block08	db7c711a6c824c42e7096d4867429bbe94cb6ffb
Final block09 / adjacent base	ad7a3fe0d46b0cd6f8595b6c23b07933113a1424
Combined source snapshot	cc19d426f17bdae34dae3c5718511d9b97ad0a7e
Validated runtime snapshot	6309633ec6eb9a6a64e8173dff949c3f8e1c5166
Block10 implementation	7e9fd56c6a0256e48fc0b1166d520c87cd1eb0cb
Final predecessor merge	ef46ceaff50b16e8e8f4817618c85a78ef74c1ab
Python companion / PR121	41af909341965c81b8afadd38f299d6f929089af
Build image	sha256:7438e53933d47d53157ed6d96d864cb208597e62dce26235ace09d1063427fa2

Working branch: audit/715-stack08-10-agent3-integration, isolated repository /private/tmp/kk715-stack10-agent3. Proposed review base is the exact block09 commit, using a pinned release/715-stack-09-base-for-10 alias when published. Neither this alias nor a new block10 PR is claimed published. Historical PR840 remains separate. The containing report commit is recorded in the live receipt.

Canonical SOP SHA256: cd2c546d41f0a723f6ce84199a3fb03661f79ab3f49fbafca08de96a7c309e89. Canonical master template SHA256: f3a49aa774f468e58af74d9bf2e4c9da83fb1082446632b85680e7b1d3c656f7. The original seven-path historical block10 reconciliation remains in 715-10-reconciliation-20260927.json and 715-10-audit-20260927.md/PDF. The tables below show net integration changes; overlapping commit counts are not added together as unique changes.

Reconciliation and findings

Block08 report authority now uses the platform PR base; push/manual runs require an independently configured accepted SHA and fail closed when absent. Exact skip reasons, name boundaries, empty waiver ledgers, shared pinned cppcheck, cache policy, portable tool discovery, real CMake test membership and explicit base ancestry checks are included. Agent1's owner acceptance and historical external verdicts remain separate; none is relabeled clean by this report.

Block09 includes the four repaired authenticator identity, consent, error and numeric-input findings, with Python41af909 and its report enforcement. The final C136357317080 passes all required jobs at the recorded adjacent base. Independently downloaded PDF and merged XML digests, wrapper source hash, identities and counts match: 1509 pass, 89 skip, zero failures/errors in the full report. Agent2's 5332680187 review has no inline findings or unresolved threads, but its receipt retains an external-acceptance caveat. Owner completion of block09 is recorded without rewriting that external verdict.

Block10 displays unknown ERC20 contract, recipient/spender and exact uint256 raw base units in the confirmation buffer. AdvancedMode, raw-data confirmation, unlimited-approval refusal, account destination binding and abort/scrub behavior are retained. Only unknown-token amount formatting gets the account-transfer fallback; other formatting failures still reject. Existing typed-array logic is preserved and two historical dimension cases are added. All74 predecessor test identities remain; two stale placeholder assertions now require raw review while preserving non-alias checks. CI runs the seven new EVM wire contracts.

INT-09-002 is retained as a separate integration repair: unchecked strtol followed by uint8 conversion allowed malformed or wrapped account-slot text to select slot0. getAuthAccount now uses the checked decimal parser with AUTHDATA_SIZE-1 before lookup, and the parser rejects NULL. Empty text, signs, spaces, suffixes, colon and overflow return Slot request out of range without account identity, Success or ButtonRequest. Valid lookup after each refusal remains unchanged. Native and real Ping tests run in both products. The old-source wire negative control fails at c48e66c4f3ec427cb2e783deef065a48f111fdce. No secret disclosure, PIN bypass, storage-format change or general64-bit HOTP support is claimed.

Git now classifies PDF artifacts as binary. This repairs whitespace preflight false positives on PDF object syntax and does not alter PDF contents.

Verification and evidence boundaries

Check	Result and scope
Runtime preservation	lib/include/unittests/deps/CMake/cmake/emulator paths compared against validated6309633ec. Only a Solana options comment differs; effective options are identical.
Full native	669 firmware +19 board +18 crypto pass.
Bitcoin-only native	143 firmware +19 board +18 crypto pass.
Full host	838 pass,84 skip,zero fail/error. Owned block08/09/10 and slot contracts execute.

Check	Result and scope
Bitcoin-only host	365 pass,557 skip,zero fail/error. Authenticator and slot execute; EVM skips are explicit product guards.
ARM/SRAM	Both links and gates pass; reserve17800B full/39136B BTC; largest frame7664B.
Merged gate regressions	All30 report, analyzer and portable preflight tests pass on final integration.
Actual evidence validator	All2730 actual full/BTC merged JUnit records accepted. Injected undeclared waiver rejected. This is a cross-variant record count, not unique coverage.
Independent authority controls	Real Git fixture rejects candidate-expanded waiver under PR, push and manual events.
OLED	Five unique maximum-approval screens inspected; contract/counterparty and all78 decimal digits visible. Prior capture carries forward because runtime is unchanged.
Local preflight	Full final invocation and outcome recorded in live readiness receipt; initial PDF classification failure retained as history.

Runtime results were executed at6309633ec with identical runtime/test/pin content; they are carried forward explicitly, not described as fresh reruns. The changed validator and preflight have fresh tests. Existing hashed100-file archive 715-08-10-local-evidence-20260927.tgz has SHA256 d4c99db7114bec9fc74cdca18d54bd96e5c2bbd47c2b775afcf3c8327e76707. It preserves raw XML, old-code controls, logs, build identity, SRAM evidence, OLED and complete skip ledgers. The final readiness evidence archive supplements that historical bundle with integration preservation, gate and predecessor checks.

Remaining skips cover deliberate BTC exclusions, five deferred capabilities, legacy endpoints, dylib suites, emulator timer limits and session power-cycle limitations. They are not passing or physical-device evidence. Both products' five storage boot/upgrade checks run with the correct emulator executable. No new runtime/pin change was introduced after the validated source. No block10 hosted artifact identity or live PR-base verification is claimed before publication.

Immutable inventories

Generation predecessor: f6da1e4e18f1ef304d5a8b5fd048dd0d93ee7285. Adjacent reproduction: git diff --numstat ad7a3fe0d46b0cd6f8595b6c23b07933113a1424..f6da1e4e18f1ef304d5a8b5fd048dd0d93ee7285. Combined reproduction: git diff --numstat 2483977523d9b6beab52a69429ace9aeefd00f76..f6da1e4e18f1ef304d5a8b5fd048dd0d93ee7285. All paths include source, tests, workflow, dependencies, source reports, binary PDFs and evidence. The final committed diff is checked for identical paths/counts.

Block10 against completed block09

Added	Deleted	Path
1	2	.gitattributes
1	1	.github/workflows/ci.yml
201	0	docs/release/audit-units/715-08-10-integration-20260927.md
-	-	docs/release/audit-units/715-08-10-integration-20260927.pdf
-	-	docs/release/audit-units/715-08-10-local-evidence-20260927.tgz
163	0	docs/release/audit-units/715-10-audit-20260927.md
-	-	docs/release/audit-units/715-10-audit-20260927.pdf
31	0	docs/release/audit-units/715-10-completion-plan-20260927.md
-	-	docs/release/audit-units/715-10-final-readiness-evidence-20260927.tgz
73	0	docs/release/audit-units/715-10-intake-20260927.md
48	0	docs/release/audit-units/715-10-reconciliation-20260927.json
4	0	include/keepkey/firmware/ethereum.h
3	7	lib/firmware/authenticator.c
53	13	lib/firmware/ethereum.c
16	9	lib/firmware/fsm_msg_ethereum.h
14	0	scripts/emulator/python-keepkey-tests.sh
36	0	scripts/emulator/test_stack09_integration.py
141	0	scripts/emulator/test_stack10_regressions.py
18	0	unittests/firmware/authenticator.cpp
64	0	unittests/firmware/eip712_stream.cpp
110	6	unittests/firmware/ethereum.cpp

Complete combined range against accepted7b

Added	Deleted	Path
1	2	.gitattributes
21	31	.github/workflows/ci.yml
3	0	.gitignore
5	0	.gitleaksignore
1	1	deps/python-keepkey
201	0	docs/release/audit-units/715-08-10-integration-20260927.md
-	-	docs/release/audit-units/715-08-10-integration-20260927.pdf
-	-	docs/release/audit-units/715-08-10-local-evidence-20260927.tgz
110	0	docs/release/audit-units/715-08-audit-20260927.md
-	-	docs/release/audit-units/715-08-audit-20260927.pdf
86	0	docs/release/audit-units/715-08-consolidated-audit-20260927.md
-	-	docs/release/audit-units/715-08-consolidated-audit-20260927.pdf
-	-	docs/release/audit-units/715-08-local-evidence-20260927.tgz
175	0	docs/release/audit-units/715-08-reconciliation-20260927.json
80	0	docs/release/audit-units/715-08B-audit-20260927.md
-	-	docs/release/audit-units/715-08B-audit-20260927.pdf
114	0	docs/release/audit-units/715-09-audit-20260927.md
-	-	docs/release/audit-units/715-09-audit-20260927.pdf
-	-	docs/release/audit-units/715-09-local-evidence-20260927.tgz
145	0	docs/release/audit-units/715-09-reconciliation-20260927.json
163	0	docs/release/audit-units/715-10-audit-20260927.md
-	-	docs/release/audit-units/715-10-audit-20260927.pdf
31	0	docs/release/audit-units/715-10-completion-plan-20260927.md
-	-	docs/release/audit-units/715-10-final-readiness-evidence-20260927.tgz
73	0	docs/release/audit-units/715-10-intake-20260927.md
48	0	docs/release/audit-units/715-10-reconciliation-20260927.json
2	0	include/keepkey/firmware/authenticator.h
4	0	include/keepkey/firmware/ethereum.h
0	1	include/keepkey/transport/messages-osmosis.options
2	3	include/keepkey/transport/messages-solana.options
113	79	lib/firmware/authenticator.c
53	13	lib/firmware/ethereum.c
1	0	lib/firmware/fsm_msg_common.h
16	9	lib/firmware/fsm_msg_ethereum.h
1	0	scripts/cppcheck-version
32	0	scripts/cppcheck.sh
14	0	scripts/emulator/python-keepkey-tests.sh
36	0	scripts/emulator/test_stack09_integration.py
141	0	scripts/emulator/test_stack10_regressions.py
96	7	scripts/generate-test-report.py
103	0	scripts/preflight.sh
92	0	scripts/preflight_checks.py

Added	Deleted	Path
248	0	scripts/test_generate_test_report.py
124	0	scripts/test_preflight_checks.py
91	0	scripts/test_preflight_cppcheck.py
214	4	unittests/firmware/authenticator.cpp
64	0	unittests/firmware/eip712_stream.cpp
110	6	unittests/firmware/ethereum.cpp

Audit handoff and remaining gates

The frozen source/PDF and complete inventories support the next final audit. Local finding dispositions and negative controls are preserved. Before spending any block10 Copilot request, verify the final adjacent code-bearing PR, exact-head hosted CI/artifact identities and final report render/preflight. The authorized budget is three initial requests, one binary split if needed, then three per unit; no requests have been used and no split has been used by agent3. Respect any later owner direction. No product merge, release promotion or physical-device acceptance is included.